

Title

Near-infrared fluorescence improves lymph node detection compared to conventional dye in breast cancer surgery

Abstract

Background: Accurate sentinel lymph node identification remains critical for breast cancer staging. Conventional blue dye techniques have limitations including inconsistent identification rates and occasional allergic reactions. We compared a novel fluorescent imaging approach to standard dye tracing. **Methods:** 40 women with node-negative early-stage breast cancer underwent sentinel lymph node biopsy using either near-infrared fluorescence imaging (n=20) or patent blue V dye (n=20). All patients then received completion axillary lymph node dissection for pathological correlation. **Results:** Near-infrared fluorescence achieved 100% sentinel node identification (20/20) versus 75% with blue dye (15/20, $p=0.047$). Median time to identification was shorter with fluorescence (14.5 min vs 24 min, $p<0.001$) and more nodes were recovered per patient (mean 3.6 vs 2.4, $p=0.002$). Sensitivity for detecting nodal metastases was 100% with fluorescence versus 75% with blue dye; false-negative rate was 0% versus 25%, respectively. No fluorescence-related adverse events occurred; blue dye caused dermal discoloration in 75% of patients and one allergic reaction. **Conclusions:** Near-infrared fluorescence provides superior identification accuracy and reduced complications compared to blue dye alone for sentinel lymph node detection in early-stage breast cancer surgery.